

Online Sales Data Analytics Project

Excel End-to-End Data Pipeline, Feature Engineering & Executive Dashboard

Project Overview & Objectives

This project demonstrates an end-to-end data analytics workflow in Microsoft Excel, transforming raw e-commerce transaction logs into a clean, feature-engineered dataset and an interactive executive dashboard.

TOTAL REVENUE

\$80,567.85

TOTAL ORDERS

239

AVG ORDER VALUE

\$335.70

Data Pipeline & Technical Execution

1. Feature Engineering & Formulas Implemented

- Month Extraction (Column K):** Extracted standard abbreviated month name from raw date timestamps.

```
=TEXT(C2, "Mmm")
```

- Customer Revenue Tiering (Column L):** Categorized transactions based on order volume threshold logic.

```
=IF(H2>=1000, "High Value", IF(H2>=200, "Medium Value", "Low Value"))
```

- Executive KPI Linking (Dashboard):** Linked dynamic summary values across sheets without hardcoding.

```
=Data!O2 (Total Revenue) | =Data!O4 (Average Order Value)
```

2. Revenue Breakdown by Category

Product Category	Total Revenue (\$)	Revenue Share (%)
Electronics	\$34,982.41	43.4%
Home Appliances	\$18,646.16	23.1%
Sports	\$14,326.52	17.8%
Clothing	\$8,128.93	10.1%
Beauty Products	\$2,621.90	3.3%
Books	\$1,861.93	2.3%

Product Category	Total Revenue (\$)	Revenue Share (%)
Grand Total	\$80,567.85	100.0%

Interactive Dashboard Architecture

The presentation layer was built on a dedicated Dashboard tab, consisting of:

- **Pivot Tables:** Grouped revenue metrics aggregated by Product Category and regional dimensions.
- **2D Clustered Column Chart:** Visual representation of category breakdown for instant visual insights.
- **Interactive Region Slicer:** Allowed dynamic filtering across *North America*, *Europe*, and *Asia* with zero latency.